# Predicting the Next Category and Region of a Visit: A Check-in-Level Multi-Task Study on Mobility Data

Vitor H. O. Silva, Germano B. dos Santos, Fabrício A. Silva
Laboratório de Inteligência em Sistemas Pervasivos e Distribuídos (NESPeD-LAB)
Universidade Federal de Viçosa, Florestal, MG, Brazil
E-mail: {vitor.h.oliveira, germano.santos, fabricio.asilva}@ufv.br

***Abstract*—Location-based social networks (LBSNs) record where people go and what they do, one check-in at a time. If we can anticipate the next visit, mobile and urban services can act ahead of demand. Two coarse questions about the next visit are usually enough: its category (the kind of place) and its region (which part of the city). These are normally handled by separate models. We therefore ask whether one model can learn both, and what sharing a single representation costs. We first build a check-in-level representation that describes each visit in its own context, instead of giving every place one fixed vector. Across five U.S. states, this lifts next-category prediction by a wide margin over a standard place embedding [1] (about $+28$ to $+40$ macro-averaged F1), and most of the gain comes from the per-visit context. We then train one multi-task model for both tasks. At every state it outperforms a dedicated category model (about $+4.7$ to $+7.7$ macro-F1), and on region it outperforms that model where regions are many, while matching it (statistically, within two points) where regions are few. The gain on region grows with the number of regions. On a non-U.S. city (Istanbul) the result holds: the joint model outperforms on category and stays within two points on region.**

***Index Terms*—next-category prediction, next-region prediction, multi-task learning, check-in-level representation, location-based social networks, mobility data**

## I. Introduction

Location-based social networks, such as Gowalla [2], let people share the places they visit. Each check-in is a small record of human movement, and together these records show how people move through a city. A natural and useful question is what a person will do next, because a service that can anticipate the next move can prepare ahead of time instead of reacting after the fact. For a mobility-aware service this can mean staging the right content in the area a user is heading to, or planning capacity there before demand arrives, and the same signal also helps recommendation and urban analysis.

Predicting the next place exactly, a single point of interest (POI), is hard, and it is often more than a service needs. Two coarser (less detailed) questions are easier to answer and usually enough: the next category, the kind of place such as food or shopping, and the next region, the part of the city a user is heading to (a neighborhood-scale unit, the census tract in the U.S. and the *mahalle* in Istanbul). These two questions pair naturally, one for intent and one for geography, and they are easier to predict accurately than a single POI while still being useful. The question we study is whether one model should learn both at once.

Sharing one representation across tasks is not free. Joint training can help one task while hurting the other, a well-known trade-off in multi-task learning (MTL): one model does several jobs at once by sharing most of its parts, and as a result the shared parameters settle on a compromise that suits neither task perfectly [3]. Our earlier work [4] observed exactly this for next-category and next-region. So the useful question is not whether MTL is good in general, but where sharing helps, where it costs, and how to share so the gains hold and the cost stays small.

We introduce two changes and measure each one carefully. First, we build a check-in-level representation: instead of one fixed vector per place, where two visits to the same coffee shop look identical, each check-in gets its own vector that carries its context (the time, the nearby places, and the recent trail). This extends hierarchical graph representations of places [1], and the infomax idea behind them [5], by adding a fourth, check-in level beneath the place, region, and city levels (Fig. 1). Per-visit context is not new on its own; what is new is obtaining it from inside this hierarchical graph representation, a choice we test directly (Section II). Second, we train one model that predicts the next category and the next region in a single forward pass, with a shared trunk that carries the semantic context and a private spatial path for the region task (Fig. 2). We deliberately evaluate on two very different settings: five U.S. states of different sizes (Gowalla) and one international city (Istanbul). This lets us see whether the findings hold across small and large, U.S. and non-U.S. data.

Our contributions are the following.

- A check-in-level representation that extends hierarchical graph infomax from the place to the individual visit. It improves next-category prediction by a large and consistent margin over a standard place embedding, about $+28$ to $+40$ points of macro-F1 (a balanced score that counts every category equally) across all five states. A controlled test shows the gain comes from the per-visit context, not from extra supervision (Table II). The margin is large because a single fixed vector per place cannot separate places that host several kinds of activity, which per-visit context recovers.

- A single model for joint next-category and next-region prediction that shares a semantic context across the two tasks while keeping a private path for the spatial one. In one forward pass it outperforms a dedicated single-task category model at every state (by about $+4.7$ to $+7.7$ points of macro-F1). On region, it outperforms where there are many regions and stays statistically non-inferior within a two-point margin (TOST, a test of equivalence) where there are few (Table III).
- An empirical account, across five states of different sizes and one international city, of how the joint gain scales with the number of regions. The category gain holds everywhere, and the region result moves monotonically: from a non-inferior match (TOST, $\pm 2$ percentage points, or pp) at the small region counts to outperforming the dedicated model at the large, with the largest state (California) showing the largest region gain (Fig. 4). The joint results on the Gowalla states are measured over five folds at a single seed.

The rest of the paper reviews background and related work (Section II), defines the problem and the two tasks (Section III), presents our method (Section IV) and experimental setup (Section V), reports results (Section VI), discusses findings and limitations (Section VII), and concludes (Section VIII).

## II. Background and Related Work

### A. From place embeddings to per-visit embeddings

A place embedding turns a POI into a vector, so that a model can reason about places by their learned geometry, where similar or nearby places sit close together. This replaces treating each place as a bare, unrelated index, an arbitrary ID number that says nothing about what the place is or where it sits. Deep Graph Infomax (DGI) [5] is a general self-supervised method for learning node representations on graphs. Applied here to a network of places, it learns such vectors by training them to tell the real place network, linked by similarity, time, and distance, from a shuffled copy. Hierarchical Graph Infomax (HGI) [1] adds geographic structure, organizing the network as place, then region, then city, so the vectors respect where places sit. Both methods produce one vector per place, so two visits to the same coffee shop, one at 8 a.m. and one at 10 p.m., look identical to the model. Contextual check-in representations instead give each visit its own vector: CTLE [6] is the closest example, learning a vector per visit by hiding parts of a user's check-in sequence and learning to fill them back in. Our representation, Check2HGI, also works at the check-in level, but it gets there differently. It keeps the hierarchical place-to-region-to-city network and trains it with the same infomax objective, now extended one level deeper, from individual places down to individual check-ins, a fourth level beneath place, region, and city (Fig. 1). It does not switch to a sequence model, as CTLE does. In short, earlier place embeddings (DGI, HGI) changed *which* features a place vector encodes; CTLE and we change *how often* a vector is produced, once per place or once per visit.

The novelty is this specific combination: per-visit context inside a hierarchical graph-infomax representation. We compare against CTLE directly to show that the combination, not per-visit context on its own, is the source of the gain.

### B. Predicting the next category and the next region

A large body of work predicts the next place a user will visit, the exact POI, with recurrent and attention models such as ST-RNN [7], DeepMove [8], Flashback [9], STAN [10], and GETNext [11]. In contrast, we target two properties of the next visit, its category and its region, rather than the exact place. These targets are not trivial: the region alone spans hundreds to several thousand classes. We choose them because they are what a mobility-aware service can act on, not to make the task easier. The category task itself follows our earlier line of work [4]. The field increasingly models several granularities at once, predicting category-level and region-level sequences alongside the place sequence; in that work, however, category and region are auxiliary signals that help a headline next-place task (SGRec [12], MCMG [13], HMT-GRN [14]). We study the pair as the object itself. To our knowledge, fine-grained region as a co-equal end target, rather than an auxiliary coarse cell, is underexplored: HMT-GRN, for instance, predicts a coarse geohash cell only as an aid to its next-place search. We therefore define our region unit carefully when we state the tasks.

### C. Multi-task learning for mobility

Multi-task models for mobility have paired location prediction with related signals for some time. MCARNN [15] jointly predicts activity and location, and iMTL [16] couples them in a shared, interactive framework built for sparse check-ins. The closest alternative to our setting is the cascade. CSLSL [17] predicts in a chain (when, then what, then where), with location as the headline output and category as an instrumental step along the way; CatDM [18] likewise uses category instrumentally, to filter candidate places. We differ in two ways. We predict category and region in parallel, as co-equal end targets in one model with a single forward pass, and we drop next-place entirely, so neither task is instrumental to a third. On the optimization side we are deliberately conservative: hard parameter sharing, one shared stack of layers for both tasks, is standard practice [3], and our use of it is not itself the contribution. Rather than propose yet another gradient-balancing method (PCGrad [19], GradNorm [20], Nash-MTL [21]), we ask where sharing helps and where it costs, and we give a clear, measured account, consistent with reports that such methods rarely improve on a well-tuned fixed weighting at two tasks [22], [23]. We confirm this on our own two tasks: across the gradient-balancing methods we tried, including the three named above, none improved on a tuned fixed task weighting in our model. The reason is visible in the gradients themselves: the next-category and next-region updates barely correlate on the shared layers (a near-zero correlation, about 0.001, throughout training), so the two tasks neither pull against each other nor reinforce each other there, and a balancer has no conflict to resolve. This is consistent with the mechanism we report in Section VI-B: because the tasks do not compete on the shared layers, the category gain comes from a stronger shared trunk rather than from the region task teaching the category one. We report this as our finding for this pair of tasks, not as a general rule. The novelty here is empirical, not architectural.

## III. Problem and Tasks

We work with check-in sequences. For each user we order their check-ins in time and form short windows of nine consecutive visits. From each window we predict two properties of the next visit. The first is its *category*, the kind of place, one of seven fixed labels: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. Istanbul uses these same seven categories through the Massive-STEPS seven-root mapping, so the category label set is identical across all datasets. The second is its *region*, the census tract the place falls in (the *mahalle* for Istanbul). We do not predict the exact next place; these two properties are easier to learn and, for most uses, enough. Category is a small, fixed label set. Region is a large one: we frame next-region as classification, where each candidate census tract is one class, from about five hundred (Istanbul) to about eight thousand five hundred (California). The spatial task is thus a fine-grained choice over many candidate regions, while the semantic task is a choice among a handful of kinds of place.

The motivation is practical, and the two predictions correspond to two kinds of preparation a mobility-aware service makes. The category says what kind of place comes next, which indicates the content or service a user will want; the region says where it is, which indicates where to prepare. With both in hand, a service can act ahead of regional demand, for example by staging content in the region a user is heading to, or by planning capacity there before the load arrives. These examples are scoped to what we actually measure. A census tract is a neighborhood, not a radio cell (the small coverage area a single base station serves); even the ten most likely tracts are far too coarse to drive cell association or handover. We therefore keep the motivation at the regional level: demand and load anticipation, content staging, and capacity planning. With that framing in place, we study how well one model can anticipate both the category and the region of the next visit. We do not build or evaluate such a service, or any network system, here; the application is the reason the predictions matter, not a result we claim.

## IV. Method

### A. The check-in-level representation

We want each visit, not each place, to carry its own vector. A standard place embedding gives every point of interest one fixed vector, so two visits to the same café look identical even when they happen at different times of day, in different company, or after different trips. We instead describe each check-in in its own context. Figure 1 shows the full data flow, from the raw check-in trail to the two outputs.

We build a graph with four levels: the check-in, the place it records, the region (a census tract) the place sits in, and the city. Edges tie these levels together and tie check-ins to one another, encoding category similarity (visits to similar kinds of place), time (visits close in time), and distance (visits close in space). This keeps the place-to-region-to-city geography of a hierarchical graph and adds the check-in as a new bottom level. We train the graph with an infomax objective, which contrasts real neighborhoods against shuffled ones, so each vector learns to match its true neighborhood and reject a fake one [1], [5]. The objective uses no task label: it never sees the next category or the next region. From the trained graph we read out one vector per check-in, and a model then sees a sequence of these per-visit vectors rather than a sequence of repeated per-place ones.

Each ingredient is individually standard: per-visit context, hierarchical place graphs, and the infomax objective all appear in prior work. The novelty is their combination, producing a vector per check-in from inside a hierarchical-graph-infomax representation, which earlier contextual embeddings built from sequence models (CTLE) do not provide. We show later that this combination, not extra supervision, is what makes the category task far easier to learn.

### B. The single multi-task model

We then train one model that reads a window of recent per-visit vectors and predicts two properties of the next visit at once: its category and its region. Figure 2 shows the design. Two input streams pass through private per-task encoders (a small input network dedicated to each task, with no weights shared between them) into a shared trunk (the stack of layers both tasks use), where the two streams exchange information (a cross-attention layer lets each stream read the other's features). The trunk then feeds two outputs, one for the category and one for the region. The region output keeps a private spatial path of its own, a small branch that the category task does not touch.

The private spatial path is a task-specific branch inside the one model, not a second model, not a second representation, and not a route that swaps models per task. The whole system is a single model: one set of shared parameters and one forward pass produce both predictions. This is the deployment property we care about, since a service that wants both answers runs a single model once, instead of two separate dedicated single-task models.

The training loss is a fixed-weight sum of the two outputs, and both outputs use plain unweighted cross-entropy. The two tasks pull in opposite directions on this choice: the region metric counts every test visit once, so an unweighted loss matches it, whereas the category metric, macro-F1, counts every class equally, so one might expect a class-weighted loss to help. We tested class-weighting on both heads and it did not help: it lowered region accuracy and category macro-F1 rather than raising it. We therefore keep plain unweighted cross-entropy on both and let the equal-weight sum balance the two tasks. This is ordinary hard parameter sharing [3], kept

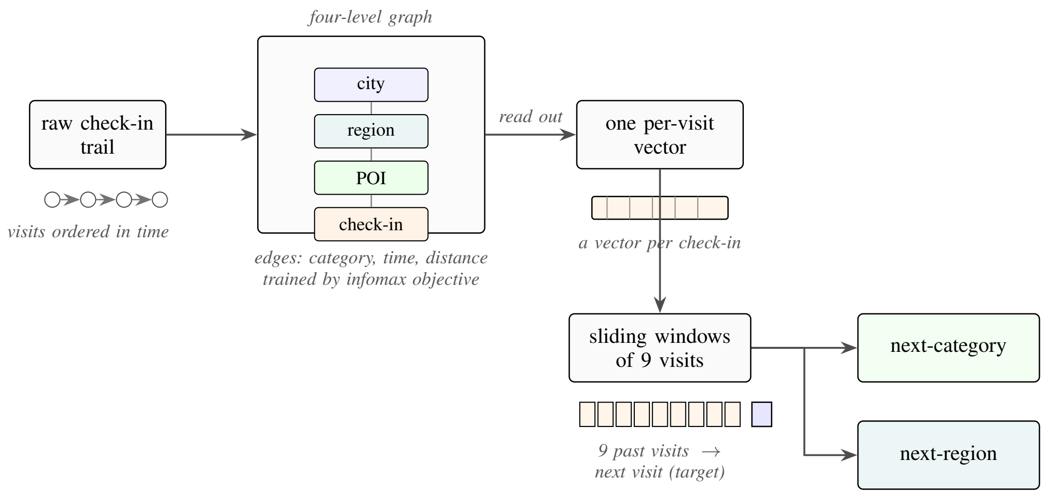


Fig. 1. From a user's check-in trail to two predictions. Each visit is placed in a four-level graph (check-in, place, region, city) and described in its own context; overlapping nine-visit windows then feed the model, which outputs the next category and the next region.

deliberately plain. Any improvement of the joint model over the dedicated single-task models then comes from the shared representation, not from a tuned weighting scheme.

The cost of serving both tasks this way is modest. The joint model carries both task heads, so it is larger than a single dedicated model. It is still substantially cheaper to serve than the two separate dedicated models the usual setup would deploy, so two answers come at well under the price of two.

## V. Experimental Setup

### A. Data

We evaluate on six datasets, summarized in Table I. Five come from Gowalla, a location-based social network, and cover the U.S. states of Alabama, Arizona, Florida, Texas, and California [2]; the sixth is the city of Istanbul, drawn from the Massive-STEPS collection [24], which gives us a non-U.S. check on the findings. We chose this spread on purpose. The states run from small to large, from about 114 thousand check-ins and 1,109 regions in Alabama to several million check-ins and 8,501 regions in California, so we can see whether a result holds as the number of regions grows. Every dataset uses the same seven place categories: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, and Travel. The region unit is a census tract for the Gowalla states and a mahalle (a municipal neighborhood) for Istanbul, and the number of regions ranges from about five hundred to about eight thousand five hundred.

### B. Windows, splitting, and the integrity of the representation

*Windows.* For each user with at least ten visits we sort the check-ins in time and form overlapping windows of nine visits, one starting at each visit, with the next visit as the target. Overlapping windows give the category task more examples to learn from. Because the windows overlap, many of a user's windows end on the same final visit, over-representing that one target; we keep a single window ending on each user's last visit and drop the duplicates. The kept window is a full-context example, not padding.

*Splitting.* We split by user with stratified five-fold cross-validation, so all of a user's windows fall in the same fold. Overlapping windows therefore cannot leak across the split, because a test user's visits never appear in training. The split is stratified on the next-category label, which preserves the class proportions across folds, and this matters because the seven-class category task is imbalanced: Food is the majority class in every dataset, from about 25 percent of visits in Florida to 34 percent in Alabama.

*Integrity of the representation.* We train the representation once on the whole dataset and then feed it to both the dedicated single-task models and the joint model. It is therefore worth being explicit that it carries no information about the test visits. It does not, for three reasons. First, the representation is trained without any task label: its objective only contrasts real graph neighborhoods against shuffled ones, and it never sees the next-category or next-region target, so no label can travel from test to train through it. Second, the only thing it could carry from the test side is graph structure (which places sit near which, in space, time, and category), because the graph is built over all places. We measured that exposure directly: for each fold we rebuilt the representation from only that fold's training users and re-ran both tasks, comparing it against the full-corpus representation on the same folds. The effect is within fold noise and shows no systematic gain from the wider exposure: on region the change is $-0.33$ points at Alabama, $+0.01$ at Arizona, and $-0.12$ at Florida. On category it is $+0.29$ at Alabama, $+0.27$ at Arizona, and $+0.00$ at Florida, measured on the in-coverage visits whose places appear in training (about 67 percent at Alabama to 87 percent at Florida); this is a place-level proxy, since the representation cannot score a place never seen in training. Third, the one component that really passes visit-to-visit information is the region-transition prior used by the spatial path: a table, built

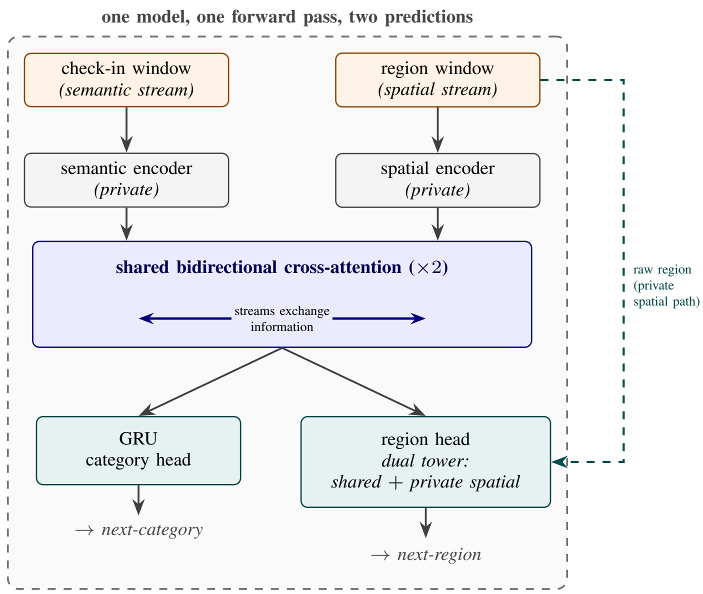


Fig. 2. The single multi-task model. Two input streams, a check-in (semantic) window and a region (spatial) window, pass through private per-task encoders into a shared cross-attention stack where each stream reads the other stream's features (the only shared parameters). A small sequence-model head reads the shared output to predict next-category; the region head combines that shared output with a private spatial path, which runs on the region window, to predict next-region. One model, one forward pass, two predictions.

from training data, of how often visits move from one region to the next, used to bias the region scores. We build it per fold from training data only, seed by seed. An earlier version that used the whole dataset inflated region accuracy by 13 to 27 points across states, which is why we now build it per fold. Every learned baseline representation is pre-trained the same way, so all inputs are held to the same standard. The one residual we cannot fully measure is visits to places never seen in training: the representation is trained on all places, so it cannot score a place that is absent from training. On the visits we can measure, which are the large majority, the audit finds no inflation, and a planned training-only variant is meant to close the remaining gap.

### *C. Metrics, superiority versus non-inferiority*

For the category task we report macro-averaged F1 (macro-F1), which averages the per-class F1 so that each of the seven categories counts equally, including the rare ones; its reference point is the majority-class floor, the macro-F1 of always predicting the single most common category. For the region task we report accuracy at ten (Acc@10), the fraction of test visits whose true region appears among the model's ten highest-scoring guesses; its comparison anchor is the dedicated single-task model.

We make two kinds of comparison and test each one appropriately. Where the joint model is meant to outperform, we test *superiority* with a paired Wilcoxon signed-rank test over the folds. Where the joint model is meant only to stay even, we test *non-inferiority*, that it is no worse than the dedicated model by more than a two-point margin, with a two-one-sided-tests procedure. We fix the two-point margin in advance and on deployment grounds, before looking at whether any claim survives it. A mobility-aware service acts on which region will be busy, not on a single rank position: it anticipates regional demand, stages content for the right region, or plans capacity at the census-tract level. A two-point shift in Acc@10 is therefore below the granularity at which such a service would behave differently. The margin is also large relative to the task: with one thousand to eight thousand five hundred regions, a random top-ten guess lands the true region under one percent of the time, so two points span a wide band of the achievable range. Finally, the equivalence is well powered. The per-fold region standard deviation is small (0.16 to 0.37 points), which gives a power near 1.0 to reject a true two-point gap, so a non-inferiority verdict reflects a real match rather than a noisy test. Which cells the joint model outperforms and which it matches is a result, reported in Section VI, not here.

### *D. Baselines*

We compare against baselines in three roles, kept separate so each answers its own question. The first role is the state of the art for each task, on our data and folds. For next-category this is POI-RGNN [25], run as a faithful author implementation, and a simple ”what usually follows what” Markov baseline over the nine recent categories (Markov-9). For next-region this is HMT-GRN [14], our primary region-native comparison, predicting the region from its own end-to-end recurrent model on the same data, seed, and folds; we run the HMT-GRN-style trunk and its region-transition prior, and drop its graph module and hierarchical beam search, which exist to prune a next-place head we do not predict. The HMT-GRN comparison is therefore a region-native model, not a component-complete reproduction. Alongside it we run a faithful STAN [10], trained from raw with its own embeddings (not ours) and its own sequence construction, and a ReHDM [26] reference reported under that method's own published protocol rather than as a matched cell. The second role is a representation control, run on Florida, to attribute the category gain to our specific design and not to contextualization in general or to extra features: CTLE [6], the closest prior contextual check-in embedding, presented fairly in both its end-to-end and frozen forms, and a feature-concatenation control that appends raw per-visit features to a standard place embedding under the same model. The third role is a multi-task comparator: the CSLSL cascade [17], which predicts in a chain rather than in parallel, so we can contrast our one-forward-pass joint model against the dominant alternative multi-task design at equal cost. The dedicated single-task models are the comparison anchors for the joint model throughout, and the standard place-level embedding [1] is the place-level comparison that isolates what the per-visit representation adds.

# VI. Results

## A. The representation makes the category task learnable

Read this as: describing each visit in its own context, rather than each place once, is what makes the next-category task learnable. Table II compares our check-in-level representation against a standard place embedding (HGI [1]) on next-category macro-F1, both fed to the same dedicated model under the windowing used throughout the paper. This is a controlled, like-for-like comparison: the target, the single-task head, and the folds are identical across the two columns, and only the input representation changes (a vector per visit versus a vector per place), so the margin isolates the representation, not a difference in task or model. The check-in-level representation outperforms the place embedding by a wide and consistent margin at every state: $+29.31$ at Alabama, $+27.63$ at Arizona, $+39.62$ at Florida, $+37.95$ at California, and $+37.47$ at Texas. The same holds on Istanbul ($+26.64$). The gains fall in two bands, about $+27$ to $+29$ points at the smaller datasets (Istanbul, Alabama, Arizona) and about $+37$ to $+40$ at the three large states. In absolute terms the place embedding recovers only about half of what the per-visit representation reaches. A controlled test attributes most of that gain to the per-visit context itself: replacing each visit's vector with a per-place-averaged version of the same representation gives back only part of the margin. This leaves roughly 64 to 90 percent of the gain (state-dependent) to the context each visit carries, not to extra training signal.

Figure 3 shows why the representation helps this task in particular. By category, the per-visit vectors are cleanly grouped. Their silhouette score by category, a separation measure from $-1$ to 1 where higher means the seven labeled groups are tighter and better separated, is about $0.56$, against about $0.00$ for the place embedding. Their nearest-neighbor category purity, the share of a vector's nearest neighbors that carry its own category, is about $0.98$, against about $0.78$. The same geometry does not separate regions, which is exactly the point: the representation improves category separability but carries no spatial structure, so its benefit is category-only and sets up the joint study in Section VI-B. A direct comparison with the closest prior contextual embedding makes the source of the gain concrete. CTLE [6], which also describes each visit but learns it with a sequence model rather than a hierarchical graph, reaches only 29.69 to 33.45 macro-F1 at Florida across its trained variants, under the same windowing and the same single-task head; this is far below our $75.15$. A feature-concat control, the place embedding joined with raw per-visit features and fed to the same model, does not close the gap either, so the category gain comes from the hierarchical per-visit representation, not from contextualization alone and not from feature injection.

## B. One model, two tasks

Read this as: one model outperforms a dedicated next-category model at every state and either outperforms or matches a dedicated next-region model, and its gain on region grows as the number of regions grows. To calibrate the two scales: category macro-F1 is taken over seven classes (a majority-class floor of about $7\%$), and region Acc@10 is taken over hundreds to thousands of regions, where a random top-ten guess is right at most about two percent of the time (and well under one percent at the larger region counts). Table III reports the single joint model against the dedicated single-task ceilings (we call them ceilings because a jointly trained head is usually expected to match or fall short of a head trained for that task alone). Figure 4 plots the signed gains ordered by region count. On category the joint model outperforms the dedicated ceiling at every state, by $+4.68$ to $+7.69$ macro-F1 (smallest at Florida, largest at Alabama, and $+6.69$ at Istanbul). On region, measured by Acc@10, the joint model outperforms the dedicated ceiling where there are many regions: Florida $+0.57$ (4,703 regions), Texas $+2.06$ (6,553), and California $+2.18$ (8,501), each with all five folds in the joint model's favor. It stays a non-inferior match (TOST, $\pm 2$ pp) where there are few: Alabama $-0.18$, Arizona $-0.06$, and Istanbul $-0.52$. The region gain increases with the number of regions, rising from $-0.52$ to $+2.18$ across the six settings, so the joint model helps the spatial task more, not less, as the number of regions grows. The state with the most regions has the largest region gain, the opposite of a cost that grows with the size of the spatial problem. We order by region count because that is the axis of interest; region count and corpus size co-vary across these datasets, so we read the trend across the six points rather than as a precise law.

The direction is unanimous: at every state all five folds favor the joint model on category, a one-sided paired Wilcoxon signed-rank test at its $n{=}5$ floor ($p{=}0.031$ per state), and the sign is consistent across all six datasets (a 6 of 6 state-level sign test, one-sided $p \approx 0.016$). At the three large states the region result likewise favors the joint model on all five folds each. We treat the state, not the fold, as the unit of replication, since folds within a state share data and the same trained models. The three small-state region results are tested equivalences (TOST, $\pm 2$ pp): Alabama ($-0.18$; 90%

TABLE I

DATASET STATISTICS, ORDERED BY NUMBER OF REGIONS. FIVE GOWALLA STATES AND ONE INTERNATIONAL CITY (ISTANBUL, FROM MASSIVE-STEPS) SPAN SMALL TO LARGE NUMBERS OF REGIONS, CHOSEN TO TEST WHETHER THE FINDINGS HOLD ACROSS SCALES AND SETTINGS. EVERY DATASET USES THE SAME SEVEN GOWALLA ROOT CATEGORIES (COMMUNITY, ENTERTAINMENT, FOOD, NIGHTLIFE, OUTDOORS, SHOPPING, TRAVEL). WINDOWS ARE THE OVERLAPPING NINE-VISIT INPUT PLUS THE NEXT VISIT AS TARGET, UNDER THE OVERLAPPING-WINDOW PROTOCOL USED THROUGHOUT; MAXIMUM AND AVERAGE SEQUENCE LENGTH ARE PER-USER CHECK-IN COUNTS. DENSITY IS THE CHECK-IN-MATRIX FILL, $100 \times$ CHECK-INS/(USERS $\times$ POIS), THE OCCUPANCY MEASURE COMMONLY REPORTED FOR NEXT-PLACE DATASETS.

| **Dataset** | **Source** | **Check-ins** | **Users** | **POIs** | **Regions** | **Windows** | **Max len** | **Avg len** | **Density (%)** |
|---|---|---|---|---|---|---|---|---|---|
| Istanbul† | Massive-STEPS | 462,615 | 23,694 | 29,816 | 520 | 270,217 | 817 | 19.5 | 0.065 |
| AL | Gowalla | 113,846 | 3,858 | 11,848 | 1,109 | 96,326 | 3,835 | 29.5 | 0.249 |
| AZ | Gowalla | 236,450 | 7,869 | 20,666 | 1,547 | 200,895 | 5,589 | 30.1 | 0.145 |
| FL | Gowalla | 1,407,034 | 21,052 | 76,544 | 4,703 | 1,274,418 | 16,679 | 66.8 | 0.087 |
| TX | Gowalla | 4,089,892 | 38,644 | 160,938 | 6,553 | 3,830,414 | 42,300 | 105.8 | 0.066 |
| CA | Gowalla | 3,171,380 | 37,090 | 169,145 | 8,501 | 2,925,466 | 14,855 | 85.5 | 0.051 |

Region unit: a census tract (a mahalle for Istanbul). † The Istanbul row uses the mahalle-level data after dropping null-coordinate visits; the raw corpus is 544,471 check-ins over 23,700 users.

confidence interval $-0.46$ to $+0.09$), Arizona ($-0.06$; $-0.41$ to $+0.29$), and Istanbul ($-0.52$; $-0.65$ to $-0.35$); all three intervals sit well inside two points. These intervals are tight because they are built on the *paired* per-fold difference, whose standard deviation is only 0.16 to 0.37 points (the per-arm fold standard deviations in Table III are larger because the two models are highly correlated across folds). The equivalence is therefore well powered. We do not claim per-cell family-wise-corrected significance at $n$=5 (a Holm correction across the six cells does not clear 0.05 at this sample size), and we leave a multi-seed confirmation ($n$=20) to future work.

A reader used to multi-task learning [3] expects the harder task to pay for the easier one, so it is worth saying where the category gain comes from. A controlled test isolates the source: we freeze the region pathway at the start of training so it cannot learn, and therefore cannot teach the category task, yet the full category lift survives at Alabama, Arizona, and Florida (within 0.3 of the joint model and far above the dedicated single-task model). We therefore read the category gain as a stronger shared trunk, obtained at no extra deployment cost (one model, one forward pass), rather than the region task teaching the category one; we report this as a finding, not a hypothesis. The joint model also stands above every external baseline on both tasks at every state (Table III): the primary region-native comparison, HMT-GRN [14] (for example California 49.61 versus our 65.66), a from-raw STAN [10] and a ReHDM reference [26] on region, and POI-RGNN [25] and a Markov floor on category, all trail it. Every region model clears a simple Markov transition floor by a wide margin. Against the closest published multi-task alternative, a cascade that predicts in a chain (when, then what, then where) [17], our parallel model reaches the same combined accuracy at the same parameter and compute cost. The difference on the joint score is about zero. We read this as a defense of the parallel design, not a claim that we outperform the cascade: the simpler parallel structure loses nothing. The Gowalla cells are seed-0 results over five folds ($n$=5, provisional); the Istanbul cells average four seeds ($n$=20).

### C. External validity (Istanbul)

Read this as: the finding holds on a non-U.S. city, read as a lift over the dedicated ceiling rather than as an absolute score. On Istanbul the joint model outperforms the dedicated category ceiling by $+6.69$ macro-F1 (53.20 to 59.89) and matches the dedicated region ceiling (TOST, $\pm 2$ pp), at $-0.52$ Acc@10 (74.80 to 74.28). The category headline is the four-seed mean against a single-seed ceiling; the clean seed-0 fold-paired gain is $+6.5$ with all five folds positive, so the result does not rest on the seed averaging. We report Istanbul as a lift over the dedicated ceiling, not as an absolute Acc@10, because its region count (520 mahalle) differs from the Gowalla states. The overlapping-window protocol, the fp32 setup, and the multi-seed run (four seeds, $n$=20) match the Gowalla board, so within Istanbul the multi-task versus dedicated comparison is like-for-like. The external baselines line up the same way as on the Gowalla states: on category the joint model is far above a nine-step Markov floor (24.55) and POI-RGNN [25] (30.12), and on region it is above a faithful STAN [10] (61.86) and HMT-GRN [14] (60.4). The picture from the U.S. states repeats on a different continent and a different region unit: the joint model outperforms on category and stays within two points on region.

## VII. DISCUSSION AND LIMITATIONS

The clearest reading of our results is a design one: one model is enough. A single model does both jobs at once. It pairs a shared trunk that carries the semantic context with a private spatial path that keeps next-region accuracy from degrading. The shared context lifts the next-category task sharply over a dedicated model at every state, and the private path keeps the next-region task competitive: a non-inferior match (TOST, $\pm 2$ pp) where there are few regions, and a clear superiority (paired Wilcoxon) where there are many. The region gain is not flat: it rises with the number of regions (Figure 4). The benefit from sharing grows as regions multiply, rather than fading. Overall, the joint model matches or outperforms the dedicated single-task model on both tasks at once (Table III). For a mobility-aware service, this points to a practical option: one model, one forward pass, can anticipate both what a user will do next and where, without paying for

TABLE II

THE CHECK-IN-LEVEL REPRESENTATION MAKES NEXT-CATEGORY PREDICTION FAR MORE LEARNABLE THAN A STANDARD PLACE EMBEDDING. A VECTOR PER VISIT, NOT A VECTOR PER PLACE, IS WHAT DRIVES THE CATEGORY GAIN. VALUES ARE NEXT-CATEGORY MACRO-F1 (MEAN AND FOLD STANDARD DEVIATION) UNDER A MATCHED SINGLE-TASK HEAD; THE CHECK-IN-LEVEL COLUMN IS THE DEDICATED CATEGORY CEILING OF TABLE III.

| State | Check-in level | Place level | Δ |
|---|---|---|---|
| Istanbul | **53.20** ±0.7 | 26.56 ±0.5 | +26.64 |
| AL | **55.87** ±2.4 | 26.56 ±1.0 | +29.31 |
| AZ | **57.13** ±1.3 | 29.50 ±1.0 | +27.63 |
| FL | **75.15** ±0.5 | 35.53 ±0.3 | +39.62 |
| TX | **69.95** ±0.2 | 32.48 ±0.6 | +37.47 |
| CA | **70.26** ±0.3 | 32.31 ±0.4 | +37.95 |

A controlled test attributes most of the gain to per-visit context: a per-place-averaged version of the same representation recovers only part of it, leaving roughly 64 to 90 percent of the gain (state-dependent) unexplained by place-level averaging (measured on the Gowalla states). The pattern holds on Istanbul as well (+26.64), so it is not a U.S.-only artifact. The matching place-level value at Alabama and Istanbul (26.56) is a coincidence of two independent runs, not a shared number. Their per-fold means differ beyond the second decimal.

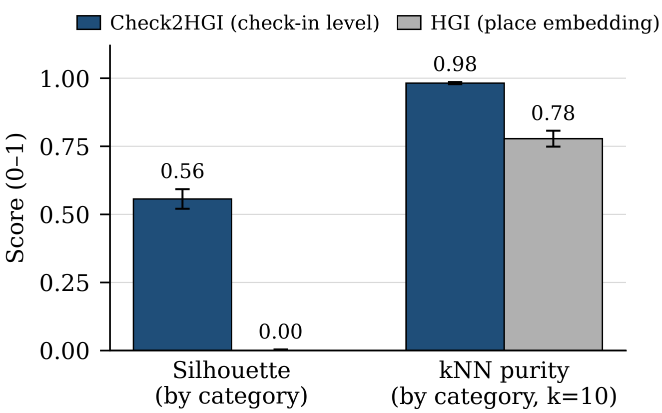


Fig. 3. Class separability of the representations, measured on the ground-truth next-category labels (cosine silhouette and leave-one-out kNN purity at $k = 10$), not on discovered clusters. The check-in-level representation cleanly separates the seven categories where the place embedding does not, which is exactly why it helps the category task and is neutral on region.

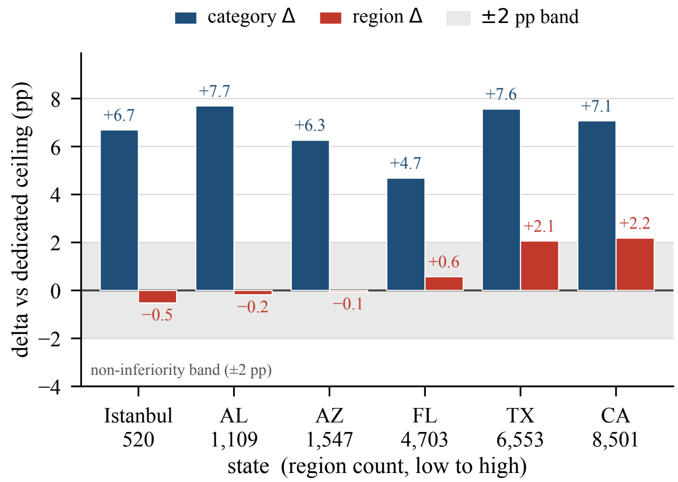


Fig. 4. Signed category and region gains of the single joint model over the dedicated single-task models, across states ordered by number of regions. The category gain (macro-F1 points) is positive at every state; the region gain (Acc@10 points) rises with region count, from within a two-point margin at the small states (AL, AZ, Istanbul) to a clear gain at the large ones (FL, TX, CA). The largest state (CA, 8,501 regions) has the largest gain. The two metrics are on different scales and are not directly comparable; the two-point band applies to the region metric. Gowalla cells are $n$=5 (seed 0) provisional; Istanbul is four-seed ($n$=20).

a second model and without trading one task off against the other.

A short usage sketch makes the motivation concrete. It remains motivation only, since we measure no service here. A service could treat the model's top-ranked next regions as an anticipatory set: a small list of areas worth preparing in advance. The next category then hints at what to prepare in them. Turning that sketch into a measured result, with its own service baselines and costs, is future work, not a claim we make.

Three limits qualify these results. First, the joint numbers come from a single seed over five folds, so they are provisional. A multi-seed run is the obvious next step and the precondition for any per-cell significance claim stronger than the ones we report. Second, our representation is trained once over all places. One slice of its behavior, visits to places never seen during training, is the single effect we cannot fully isolate. A planned variant that trains on each fold's training places only is meant to close that gap. Third, although a mobility-aware service is our motivation throughout, we do not build or evaluate one; tying these predictions to a concrete service is the natural next study.

Finally, because our representation is a fixed per-visit vector that is computed once and left unchanged, it is not tied to this particular model: any downstream model can consume it. Plugging it into other sequence models, and applying it to mobility tasks beyond next-category and next-region, are natural directions to explore.

## VIII. CONCLUSION

We asked whether one model can predict both the category and the region of a user's next visit. Two findings answer it. First, a check-in-level representation, where each visit gets its own vector instead of one fixed vector per place, makes the next-category task far more learnable than the standard place-level representation does. Second, on top of that representation, a single multi-task model outperforms a dedicated category model at every state, and on region it does so where there are many regions while remaining non-inferior (TOST, ±2 pp) where there are few. In summary, one model that shares a semantic context across both tasks while keeping a private spatial path for region can anticipate both what a user will do next and where, without giving up the next-region task. The gain on region grows with the number of regions.

## REFERENCES


[1] W. Huang, D. Zhang, G. Mai, X. Guo, and L. Cui, "Learning urban region representations with POIs and hierarchical graph infomax," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 196, pp. 134–145, 2023.

TABLE III

ONE MODEL, TWO TASKS. A SINGLE JOINT MODEL OUTPERFORMS THE DEDICATED NEXT-CATEGORY CEILING AT EVERY STATE, AND ON NEXT-REGION IT OUTPERFORMS THE DEDICATED CEILING WHERE THERE ARE MANY REGIONS (FL, TX, CA) AND IS STATISTICALLY NON-INFERIOR WITHIN A TWO-POINT MARGIN (TOST) WHERE THERE ARE FEW (AL, AZ, ISTANBUL). CATEGORY IS MACRO-F1, REGION IS ACC@10; OUR CELLS SHOW THE MEAN AND STANDARD DEVIATION OVER FIVE FOLDS. BOLD MARKS AN IMPROVEMENT OVER THE DEDICATED CEILING; ↑ MARKS THAT THE JOINT MODEL OUTPERFORMS IT (PAIRED WILCOXON); ≈ MARKS A NON-INFERIOR MATCH (TOST). STATES ARE ORDERED BY REGION COUNT.

| | | Next-category (macro-F1) | | | | Next-region (Acc@10) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| **State** | **Regions** | **Markov-9** | **POI-RGNN** | **Dedicated** | **Joint (ours)** | **HMT-GRN§** | **ReHDM** | **STAN‡** | **Dedicated** | **Joint (ours)** |
| Istanbul | 520 | 24.55 | 30.12 | 53.20 ±0.7 | **59.89** ±0.6 | 60.4 | – | 61.86 | 74.80 ±0.6 | 74.28 ±0.7≈ |
| AL | 1,109 | 20.50 | 23.80 | 55.87 ±2.4 | **63.56** ±2.0 | 57.05 | 66.06 | 60.72 | 69.99 ±3.2 | 69.81 ±3.4≈ |
| AZ | 1,547 | 23.92 | 27.64 | 57.13 ±1.3 | **63.39** ±1.3 | 43.70 | 54.65 | 49.86 | 59.40 ±1.9 | 59.34 ±1.8≈ |
| FL | 4,703 | 29.74 | 34.49 | 75.15 ±0.5 | **79.82** ±0.5 | 63.74 | 65.68 | 72.99 | 76.71 ±1.0 | **77.28** ±0.8↑ |
| TX | 6,553 | 28.67 | 33.03 | 69.95 ±0.2 | **77.51** ±0.1 | 53.85 | –† | –† | 64.96 ±0.5 | **67.02** ±0.5↑ |
| CA | 8,501 | 27.58 | 31.78 | 70.26 ±0.3 | **77.33** ±0.2 | 49.61 | –† | –† | 63.48 ±0.3 | **65.66** ±0.3↑ |

Region externals by comparability: HMT-GRN§ (region-native, on our footing) is the primary comparison; STAN (faithful, from raw) and ReHDM (its own protocol) are secondary references. The joint model is above all three at every state, and every region model clears a Markov-1 floor by a wide margin. †Faithful STAN and ReHDM are infeasible at the CA/TX region scale; STAN reports AL, AZ, FL, and Istanbul. §HMT-GRN-style: its own recurrent trunk and a train-only region-transition prior, with the graph module and hierarchical search dropped. Baseline cells are 5-fold means. Gowalla joint-model cells are $n$=5 (seed 0) provisional; Istanbul cells are four-seed means ($n$=20) against a single-seed ceiling. A cascade comparator (CSLSL) matches our parallel model at equal cost (see text).


[2] E. Cho, S. A. Myers, and J. Leskovec, “Friendship and mobility: User movement in location-based social networks,” in *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pp. 1082–1090, ACM, 2011.

[3] R. Caruana, “Multitask learning,” *Machine Learning*, vol. 28, no. 1, pp. 41–75, 1997.

[4] V. H. O. Silva, I. F. Almeida, T. S. Paiva, G. B. Santos, F. A. Silva, and F. T. Sousa, “An investigation into multi-task learning for point-of-interest category classification and next-POI prediction,” in *Anais do Congresso Brasileiro de Inteligência Computacional (CBIC)*, 2025.

[5] P. Veličković, W. Fedus, W. L. Hamilton, P. Liò, Y. Bengio, and R. D. Hjelm, “Deep graph infomax,” in *Proceedings of the 7th International Conference on Learning Representations (ICLR)*, 2019.

[6] Y. Lin, H. Wan, S. Guo, and Y. Lin, “Pre-training context and time aware location embeddings from spatial-temporal trajectories for user next location prediction,” in *Proceedings of the 35th AAAI Conference on Artificial Intelligence*, vol. 35, pp. 4241–4248, 2021.

[7] Q. Liu, S. Wu, L. Wang, and T. Tan, “Predicting the next location: A recurrent model with spatial and temporal contexts,” in *Proceedings of the 30th AAAI Conference on Artificial Intelligence*, pp. 194–200, 2016.

[8] J. Feng, Y. Li, C. Zhang, F. Sun, F. Meng, A. Guo, and D. Jin, “DeepMove: Predicting human mobility with attentional recurrent networks,” in *Proceedings of the 2018 World Wide Web Conference (WWW)*, pp. 1459–1468, 2018.

[9] D. Yang, B. Fankhauser, P. Rosso, and P. Cudré-Mauroux, “Location prediction over sparse user mobility traces using RNNs: Flashback in hidden states!,” in *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 2184–2190, 2020.

[10] Y. Luo, Q. Liu, and Z. Liu, “STAN: Spatio-temporal attention network for next location recommendation,” in *Proceedings of the Web Conference 2021 (WWW)*, pp. 2177–2185, 2021.

[11] S. Yang, J. Liu, and K. Zhao, “GETNext: Trajectory flow map enhanced transformer for next POI recommendation,” in *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 1144–1153, 2022.

[12] Y. Li, T. Chen, Y. Luo, H. Yin, and Z. Huang, “Discovering collaborative signals for next POI recommendation with iterative Seq2Graph augmentation,” in *Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 1491–1497, 2021.

[13] Z. Sun, Y. Lei, L. Zhang, C. Li, Y.-S. Ong, and J. Zhang, “A multi-channel next POI recommendation framework with multi-granularity check-in signals,” *ACM Transactions on Information Systems*, vol. 42, no. 1, pp. 1–28, 2024.

[14] N. Lim, B. Hooi, S. Ng, Y. L. Goh, R. Weng, and R. Tan, “Hierarchical multi-task graph recurrent network for next POI recommendation,” in *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 1133–1143, 2022.

[15] D. Liao, W. Liu, Y. Zhong, J. Li, and G. Wang, “Predicting activity and location with multi-task context aware recurrent neural network,” in *Proceedings of the 27th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 3435–3441, 2018.

[16] L. Zhang, Z. Sun, J. Zhang, Y. Lei, C. Li, Z. Wu, H. Kloeden, and F. Klanner, “An interactive multi-task learning framework for next POI recommendation with uncertain check-ins,” in *Proceedings of the 29th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 3551–3557, 2020.

[17] Z. Huang, S. Xu, M. Wang, H. Wu, Y. Xu, and Y. Jin, “Human mobility prediction with causal and spatial-constrained multi-task network,” *EPJ Data Science*, vol. 13, no. 1, p. 22, 2024.

[18] F. Yu, L. Cui, W. Guo, X. Lu, Q. Li, and H. Lu, “A category-aware deep model for successive POI recommendation on sparse check-in data,” in *Proceedings of The Web Conference 2020 (WWW)*, pp. 1264–1274, 2020.

[19] T. Yu, S. Kumar, A. Gupta, S. Levine, K. Hausman, and C. Finn, “Gradient surgery for multi-task learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020.

[20] Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Rabinovich, “GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks,” in *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pp. 794–803, 2018.

[21] A. Navon, A. Shamsian, I. Achituve, H. Maron, K. Kawaguchi, G. Chechik, and E. Fetaya, “Multi-task learning as a bargaining game,” in *Proceedings of the 39th International Conference on Machine Learning (ICML)*, pp. 16428–16446, 2022.

[22] D. Xin, B. Ghorbani, A. Garg, O. Firat, and J. Gilmer, “Do current multi-task optimization methods in deep learning even help?,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022.

[23] V. Kurin, A. D. Palma, I. Kostrikov, S. Whiteson, and M. P. Kumar, “In defense of the unitary scalarization for deep multi-task learning,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 35, 2022.

[24] W. Wongso, H. Xue, and F. D. Salim, “Massive-steps: Massive semantic trajectories for understanding POI check-ins — dataset and benchmarks.” arXiv:2505.11239, 2025. arXiv preprint.

[25] C. G. S. Capanema, G. S. de Oliveira, F. A. Silva, T. R. de Moura Braga Silva, and A. A. F. Loureiro, “Combining recurrent and graph neural networks to predict the next place’s category,” *Ad Hoc Networks*, vol. 138, p. 103016, 2023.

[26] X. Li, Z. Gu, R. Yao, Y. Zhou, H. Zhu, J. Zhao, and W. Du, “Beyond individual and point: Next POI recommendation via region-aware dynamic hypergraph with dual-level modeling,” in *Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI)*, pp. 3081–3089, 2025.